Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years building **React** web apps, Python and Java backends, **data pipelines** over large time-series datasets, and **dashboards** that automate manual workflows for engineering and operations users.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data pipelines** with incremental loads, validation, and reconciliation across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs at 99%+ reliability through CI/CD-backed releases for 500+ users.
- Cut **dashboard** latency from 1.2s to 350ms for 500+ users by redesigning schemas and indexed queries, and tripled async job throughput across 1K+ jobs with failure isolation.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack **React** and Flask web application on AWS that automated a manual grading workflow for 150+ instructors, handling 10K+ daily requests and cutting turnaround 60%.
- Shipped upload, rubric, and grading-history features with secure PDF, DOCX, and ZIP ingestion and multithreaded **batch processing** for 1K+ files per run, keeping APIs stable at peak load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines processing 100K+ vehicle trajectory logs sampled at 10Hz, aligning multi-source **time-series** signals into a shared frame for training and evaluating a PyTorch motion planner in simulation.
- Analyzed closed-loop **test runs** of a new encoder against three baselines, documenting failure modes such as off-road trajectories and improving long-horizon planning 18% and closed-loop stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned AWS Glue and S3 **ETL** that normalized JSON, CSV, and Parquet data from multiple sources, boosting pipeline efficiency 68% and cutting storage costs 15% for downstream analytics.
- Built object-oriented Java and Python modules covered by 20+ automated tests in nightly Jenkins **CI/CD**, and cut manual data entry 35% through automated data-sharing workflows across three platforms.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a React and Node.js **resource management** platform for resource allocation, assignment conflicts, billing, and customer records, syncing 2M+ records across sites with real-time WebSocket updates for operations teams.
- Analyzed usage and load-time data to guide fixes that cut resource wastage 42%, and shipped through Jenkins CI/CD with Docker packaging and automated end-to-end tests across release cycles.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++
Full Stack: React, Next.js, Node.js, Flask, FastAPI, Spring Boot, REST APIs, WebSockets
Data: PostgreSQL, SQL Server, MySQL, MongoDB, ETL Pipelines, Spark, AWS Glue, Time-Series Data, Tableau
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Linux
Practices: Data Structures, Algorithms, Unit Testing, PyTest, Jest, Selenium, Agile, Git, Debugging

Projects

Test Stand Telemetry Dashboard | *Python, Kafka, TimescaleDB, React, Grafana*

- Designed a pipeline that streams **test stand telemetry** such as pressures and temperatures into TimescaleDB, flags readings outside limits automatically, and shows live and replayed runs in a **React** dashboard.

Production Planning Tracker | *TypeScript, React, Node.js, PostgreSQL*

- Designed a **production planning** app that tracks parts from raw material through build steps, highlights schedule risks from late **supplier** deliveries, and lets operators update status from the floor.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Shipped a **full-stack** Next.js and TypeScript product end to end with real-time updates, transaction filtering, and Kubernetes CI/CD, supporting 50,000+ users and 1M+ transactions with MongoDB storage.